



Figure 1: System stack with EFI

In addition, while conventional boot-loaders will perhaps fail to make the cut once UEFI comes into play, alternatives such as WUBI might still go strong.

And, on a slightly more radical note, chances are that UEFI's digital keys will reject many restore-and-recovery tools too. Obviously, GNU/Linux users will not be the only ones suffering—or complaining.

There is also speculation that given the gigantic success of Android (read: gigantic failure of Windows Mobile), Microsoft might employ this secure booting concept to seize a considerable portion of tablets and other portable devices for its Mobile OS.

Any reactions yet?

To begin with, hacking or jail-breaking the UEFI will not be possible, as in all probability, Microsoft intends to get the digital signatures copyrighted. Google Chromebooks also comes with such secure booting options, but they can be disabled under Developer Mode.

Along similar lines, One Laptop Per Child (OLPC) devices are also boot-protected by default, but the protection can be eliminated by requesting a unique key, and then running the firmware command 'disable-security'. The logic behind the OLPC mechanism is to prevent the theft of laptops from children.

In such a case, Microsoft's act can by no means be called ethical. Linux Australia is planning to appeal to the Australian Competition and Consumer Commission (ACCC) claiming that the digital signatures in UEFI are non-competitive, and are a ploy to establish monopoly control over the market by unethical methods.

Also, going beyond the FOSS community, this move has not been welcomed by OEMs and hardware vendors either, as they will be at the receiving end of the end users' angry reactions to their inability to boot other operating systems.

The community response

Both Red Hat and Canonical are members of the UEFI forums. Thus, they are equally aware of the outcome if Microsoft has its way. Tactically, this might not matter much, as Microsoft commands both the financial and strategic resources to influence the entire forum.

Technically, on the other hand, Linux already supports both UEFI and BIOS firmware, while Microsoft Windows supports



Figure 2: UEFI boot-up

UEFI only in its newer releases. Clearly, the open source community is ahead. Of course, there exist certain bugs, but most of them will be patched with the release of kernel 3.0.

As mentioned above, the GPL of GRUB might not allow the inclusion of proprietary code in the boot-loader itself. Certain distros are considering abandoning GRUB (GPL licensed) and migrating to LILO (BSD licensed) since the BSD licence allows the inclusion of non-free code, and thus, LILO can be considered a worthy option (though nearly overlooked, it still continues to be in active development).

Another easier, though slightly less secure option could be to invent a public access key for all boot-loaders.

To sum it up...

Most bloggers and Internet users are speculating about the results of these stealthy war games being played around UEFI. In any case, if the worst happens, rest assured that the community will indeed figure a way out. After all, most Ubuntu users have learnt to use the Windows logo key on their keyboards to bring up the Unity sidebar, haven't they? As of now, all we can do is wait and watch!

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